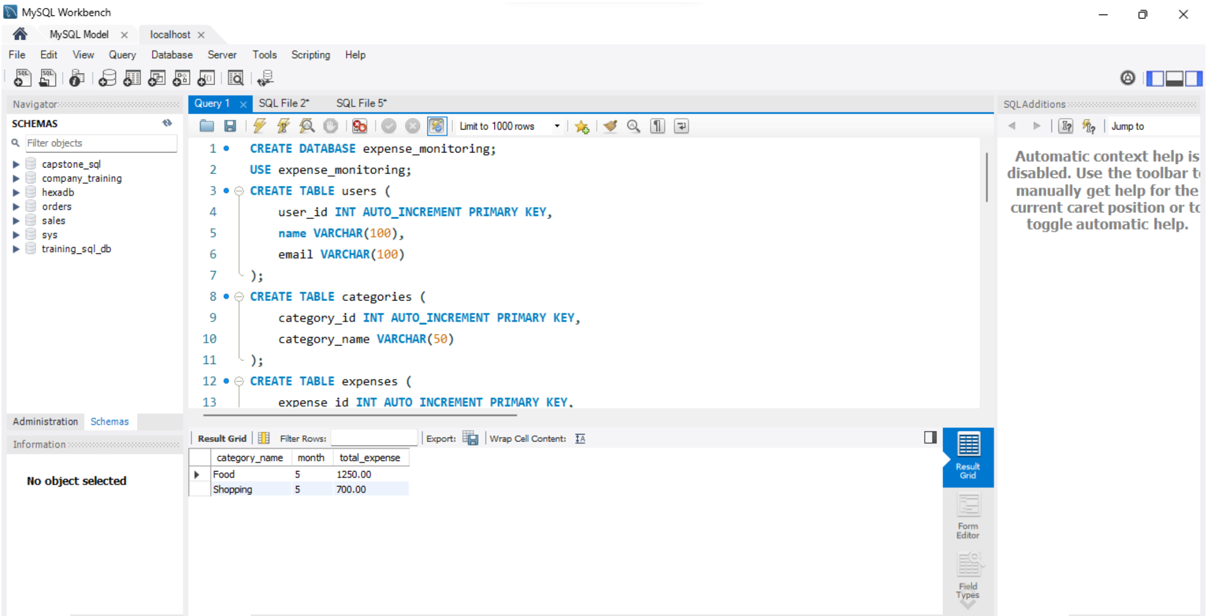
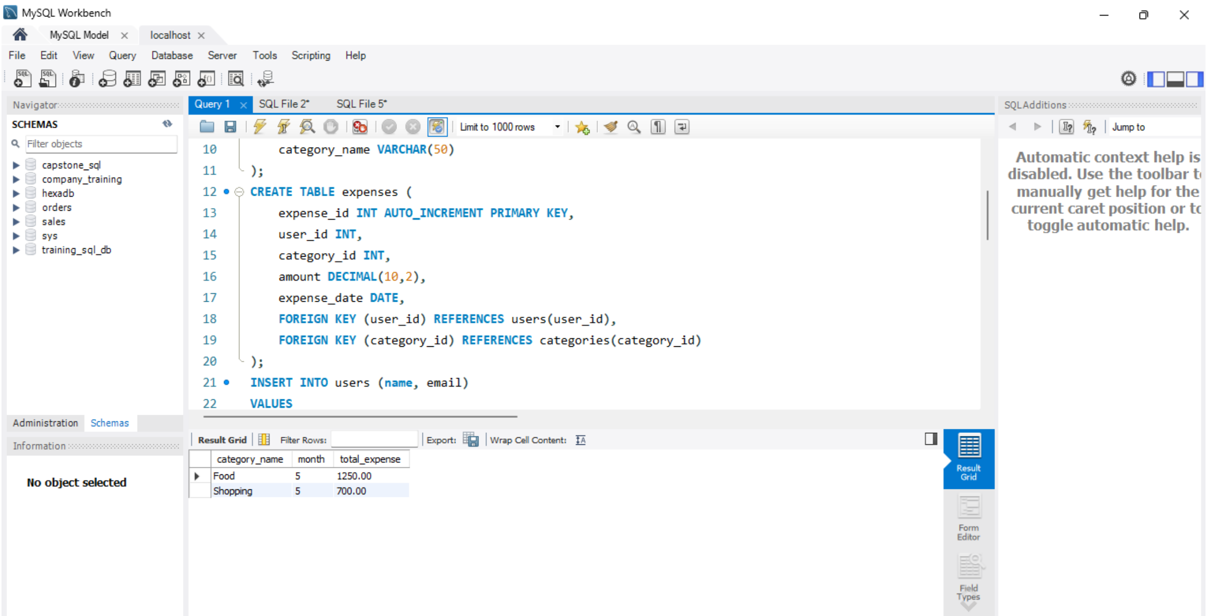
```sql
CREATE DATABASE expense_monitoring;

USE expense_monitoring;

CREATE TABLE users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100)
);

CREATE TABLE categories (
    category_id INT AUTO_INCREMENT PRIMARY KEY,
    category_name VARCHAR(50)
);

CREATE TABLE expenses (
    expense_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    category_id INT,
    amount DECIMAL(10,2),
    expense_date DATE,
    FOREIGN KEY (user_id) REFERENCES users(user_id),
    FOREIGN KEY (category_id) REFERENCES categories(category_id)
);

INSERT INTO users (name, email)
VALUES
('Thamarai', 'thamarai@gmail.com'),
('Ravi', 'ravi@gmail.com');

INSERT INTO categories (category_name)
VALUES
('Food'),
('Travel'),
```

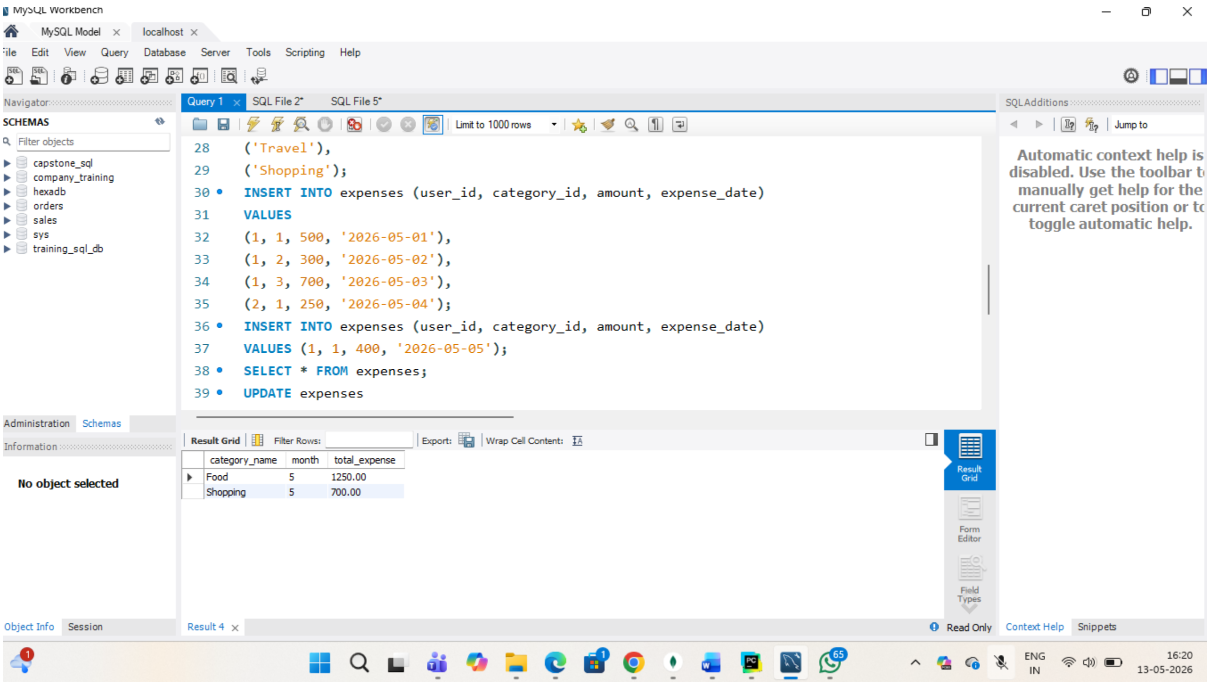
```
('Shopping');
INSERT INTO expenses (user_id, category_id, amount, expense_date)
VALUES
(1, 1, 500, '2026-05-01'),
(1, 2, 300, '2026-05-02'),
(1, 3, 700, '2026-05-03'),
(2, 1, 250, '2026-05-04');
INSERT INTO expenses (user_id, category_id, amount, expense_date)
VALUES (1, 1, 400, '2026-05-05');
SELECT * FROM expenses;
UPDATE expenses
SET amount = 600
WHERE expense_id = 1;
DELETE FROM expenses
WHERE expense_id = 2;
DELIMITER //

CREATE PROCEDURE MonthlyExpenseByCategory()
BEGIN
  SELECT c.category_name,
      MONTH(e.expense_date) AS month,
      SUM(e.amount) AS total_expense
  FROM expenses e
  JOIN categories c
   ON e.category_id = c.category_id
  GROUP BY c.category_name, MONTH(e.expense_date);
END //
```

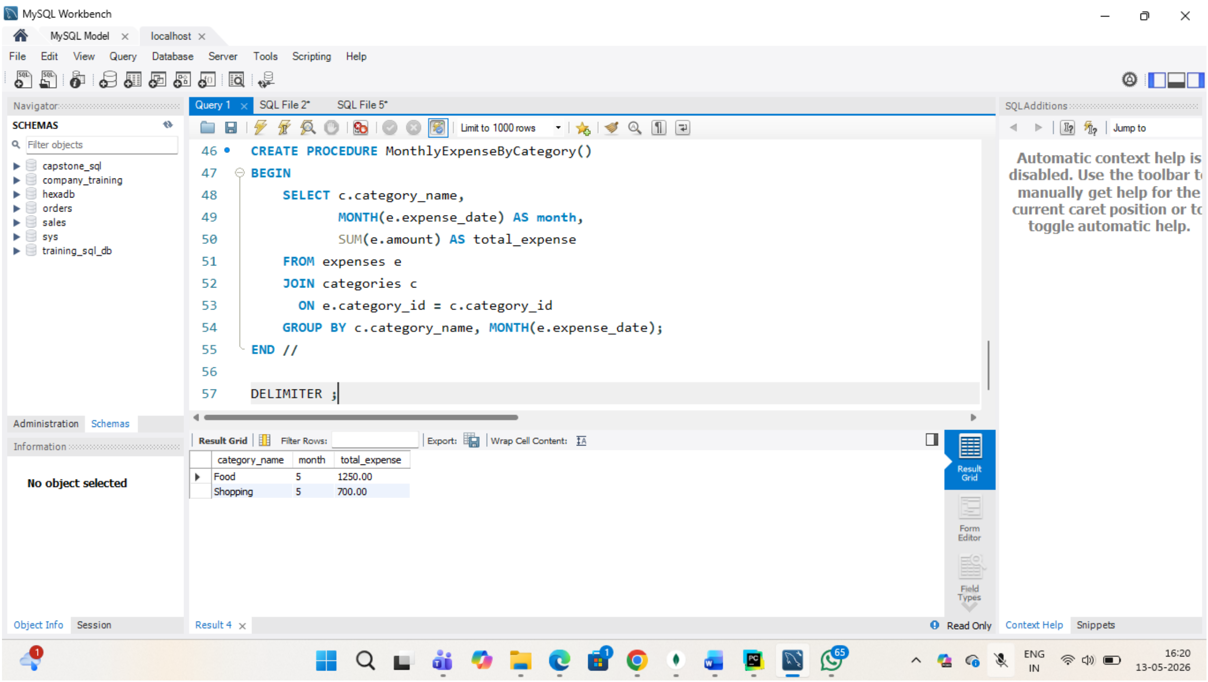
DELIMITER ;

CALL MonthlyExpenseByCategory();






MySQL Workbench
('Travel'),
('Shopping');
INSERT INTO expenses (user_id, category_id, amount, expense_date)
VALUES
(1, 1, 500, '2026-05-01'),
(1, 2, 300, '2026-05-02'),
(1, 3, 700, '2026-05-03'),
(2, 1, 250, '2026-05-04');
INSERT INTO expenses (user_id, category_id, amount, expense_date)
VALUES (1, 1, 400, '2026-05-05');
SELECT * FROM expenses;
UPDATE expenses
category_name month total_expense
Food 5 1250.00
Shopping 5 700.00
No object selected
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.


MySQL Workbench
CREATE PROCEDURE MonthlyExpenseByCategory()
BEGIN
SELECT c.category_name,
MONTH(e.expense_date) AS month,
SUM(e.amount) AS total_expense
FROM expenses e
JOIN categories c
ON e.category_id = c.category_id
GROUP BY c.category_name, MONTH(e.expense_date);
END //
DELIMITER ;
category_name month total_expense
Food 5 1250.00
Shopping 5 700.00
No object selected
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.